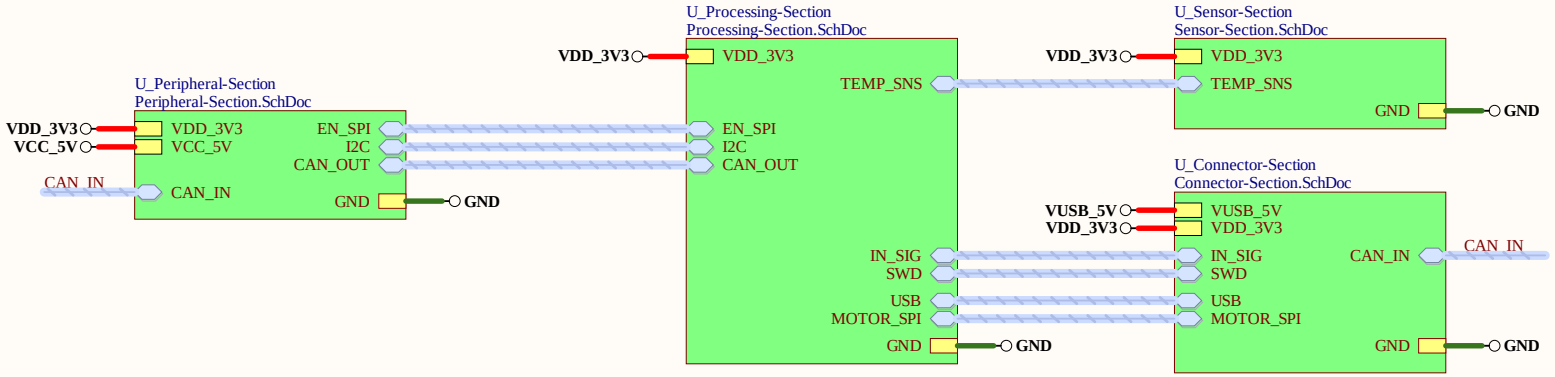
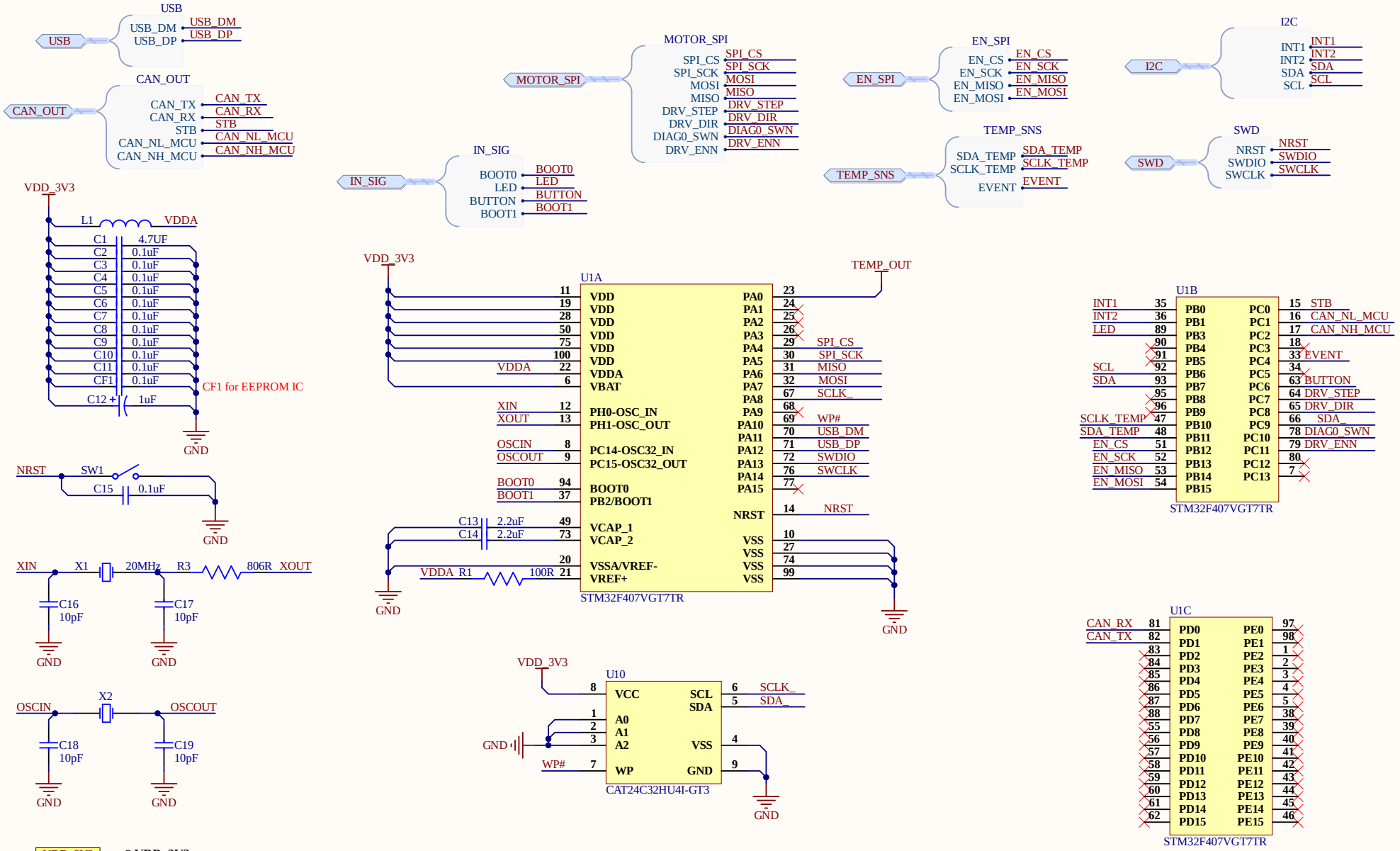
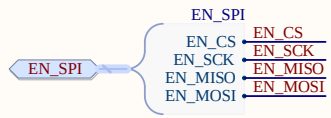
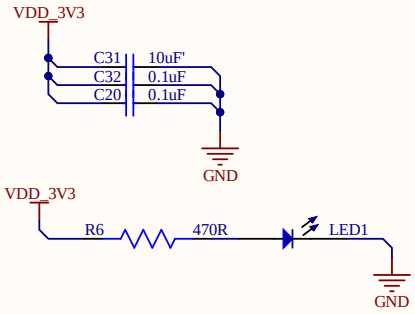
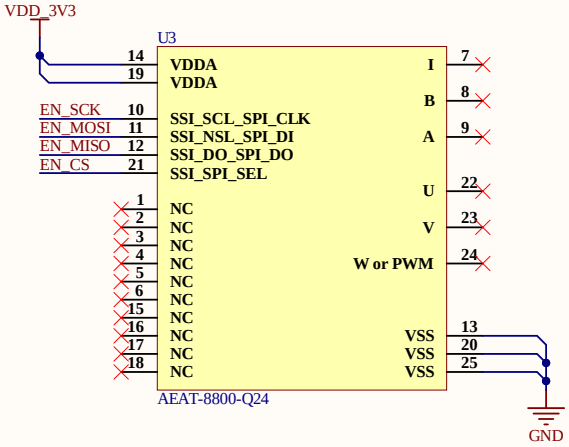
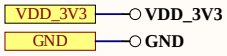

U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
EN_SPI
I2C
CAN_OUT
GND
GND
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
EN_SPI
I2C
CAN_OUT
IN_SIG
SWD
USB
MOTOR_SPI
GND
GND
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
U_Connector-Section
Connector-Section.SchDoc
VUSB_5V
VDD_3V3
VUSB_5V
VDD_3V3
IN_SIG
SWD
USB
MOTOR_SPI
CAN_IN
CAN_IN
GND
GND

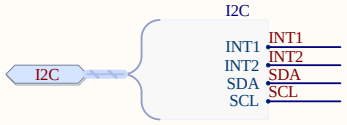

USB
USB_DM
USB_DP
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
EN_SPI
EN_CS
EN_SCK
EN_MISO
EN_MOSI
I2C
INT1
INT2
SDA
SCL
IN_SIG
BOOT0
LED
BUTTON
BOOT1
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SWD
NRST
SWDIO
SWCLK
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
CF1 for EEPROM IC
C12 1uF
GND
NRST
SW1
C15 0.1uF
XIN
X1
20MHz
R3
806R
XOUT
C16 10pF
C17 10pF
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
U1A
VDD
VDDA
VBAT
PH0-OSC_IN
PH1-OSC_OUT
PC14-OSC32_IN
PC15-OSC32_OUT
BOOT0
PB2/BOOT1
VCAP_1
VCAP_2
VSSA/VREF-
VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NRST
VSS
C13 2.2uF
C14 2.2uF
VDDA_R1 100R
STM32F407VGT7TR
TEMP_OUT
U1B
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
U1C
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
U10
VCC
A0
A1
A2
WP
SCL
SDA
VSS
GND
WP#
CAT24C32HU4I-GT3
SCLK_
SDA_
VDD_3V3
VDD_3V3
GND
GND

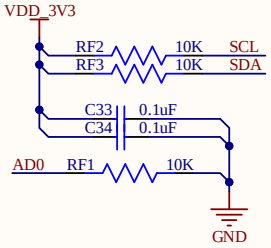

EN_SPI
EN_SPI
EN_CS
EN_SCK
EN_MISO
EN_MOSI
EN_CS
EN_SCK
EN_MISO
EN_MOSI

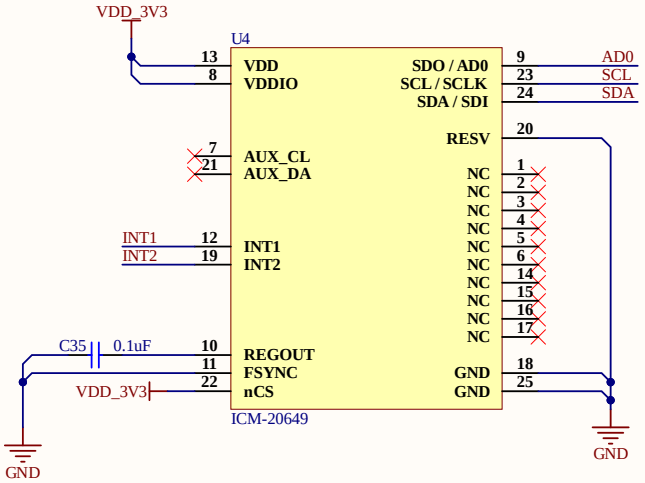

VDD_3V3
C31
10uF'
C32
0.1uF
C20
0.1uF
GND
VDD_3V3
R6
470R
LED1
GND

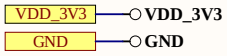

VDD_3V3
U3
14
19
VDDA
VDDA
EN_SCK
EN_MOSI
EN_MISO
EN_CS
10
11
12
21
SSI_SCL_SPI_CLK
SSI_NSL_SPI_DI
SSI_DO_SPI_DO
SSI_SPI_SEL
1
2
3
4
5
6
15
16
17
18
NC
NC
NC
NC
NC
NC
NC
NC
NC
I
B
A
U
V
W or PWM
VSS
VSS
VSS
7
8
9
22
23
24
13
20
25
AEAT-8800-Q24
GND

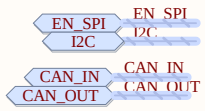

VDD_3V3
VDD_3V3
GND
GND

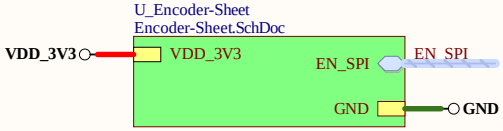

I2C
I2C
INT1
INT2
SDA
SCL
INT1
INT2
SDA
SCL

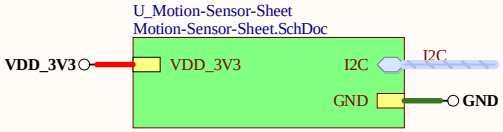

VDD_3V3
RF2
10K
SCL
RF3
10K
SDA
C33
0.1uF
C34
0.1uF
AD0
RF1
10K
GND

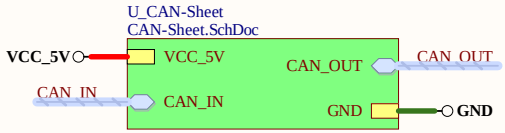

VDD_3V3
U4
13
VDD
8
VDDIO
7
AUX_CL
21
AUX_DA
INT1
12
INT1
INT2
19
INT2
C35
0.1uF
10
REGOUT
11
FSYNC
VDD_3V3
22
nCS
GND
SDO / AD0
SCL / SCLK
SDA / SDI
RESV
NC
GND
9
AD0
23
SCL
24
SDA
20
1
2
3
4
5
6
14
15
16
17
18
25
ICM-20649
GND

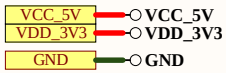

VDD_3V3
VDD_3V3
GND
GND

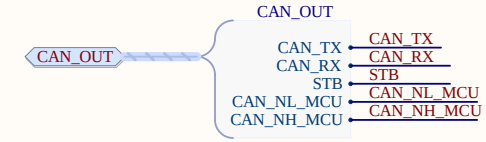

EN_SPI
I2C
EN_SPI
I2C
CAN_IN
CAN_OUT
CAN_IN
CAN_OUT

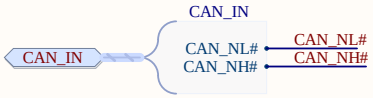

U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND

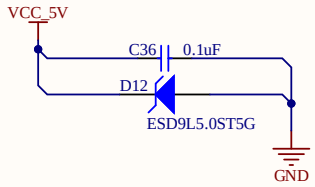

U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND


U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
GND
GND


VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND

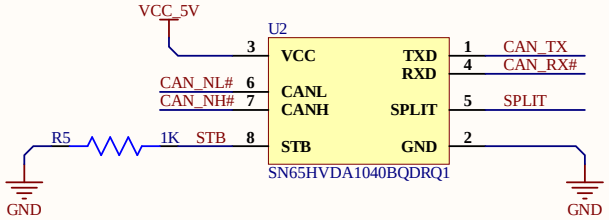


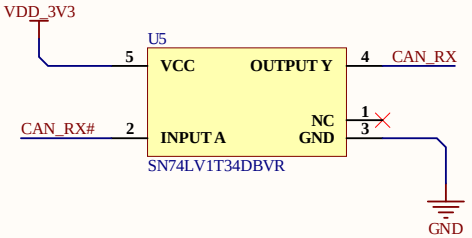


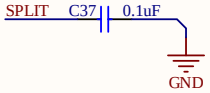


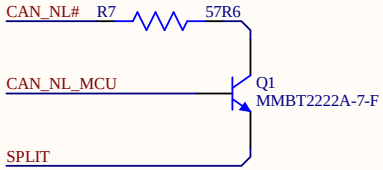


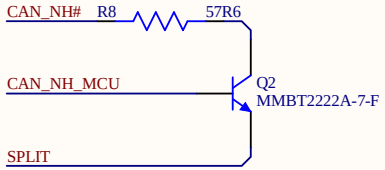
Connections for CAN are w.r.t Controller Rx and Tx

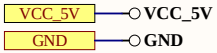


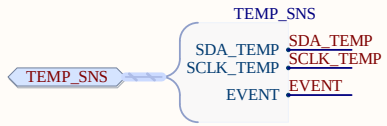


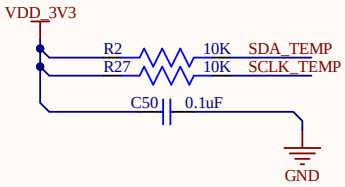


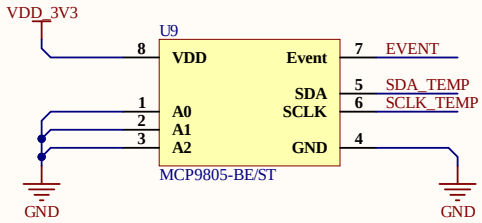


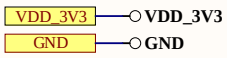


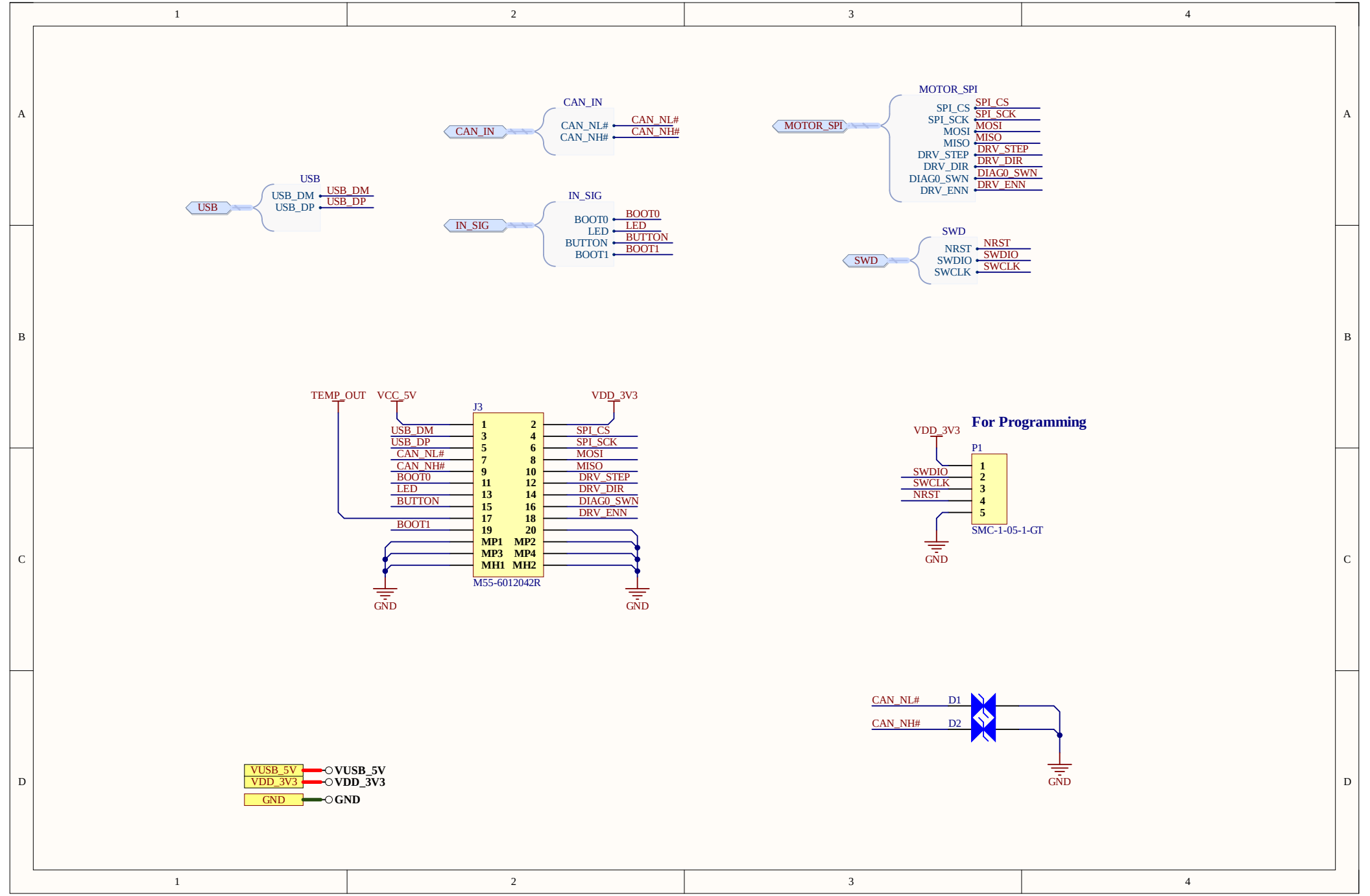



TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT


VDD_3V3
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND


VDD_3V3
U9
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND


VDD_3V3
VDD_3V3
GND
GND


1
2
3
4
A
B
C
D
CAN_IN
CAN_IN
CAN_NL#
CAN_NH#
CAN_NL#
CAN_NH#
MOTOR_SPI
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
USB
USB
USB_DM
USB_DP
USB_DM
USB_DP
IN_SIG
IN_SIG
BOOT0
LED
BUTTON
BOOT1
BOOT0
LED
BUTTON
BOOT1
SWD
SWD
NRST
SWDIO
SWCLK
NRST
SWDIO
SWCLK
TEMP_OUT
VCC_5V
VDD_3V3
J3
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
MP1
MP2
MP3
MP4
MH1
MH2
USB_DM
USB_DP
CAN_NL#
CAN_NH#
BOOT0
LED
BUTTON
BOOT1
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
M55-6012042R
GND
GND
For Programming
VDD_3V3
P1
SWDIO
SWCLK
NRST
1
2
3
4
5
SMC-1-05-1-GT
GND
CAN_NL#
CAN_NH#
D1
D2
GND
VUSB_5V
VDD_3V3
GND
VUSB_5V
VDD_3V3
GND